Complete at least 2 exercises.

Throughout, $\Bbbk$ is a field and all modules are graded right modules. A graded algebra $A = \bigoplus_{i \geq 0} A_i$ is locally finite-dimensional with $S = A_0$ semisimple; in all explicit examples $S = \bigoplus_i \Bbbk e_i$ with coordinate idempotents e_i.

Exercise 1.

(1) Let $S = \bigoplus_{i=1}^{3} \Bbbk e_i$ and let $V = \Bbbk a_1 \oplus \Bbbk a_2 \oplus \Bbbk b_1 \oplus \Bbbk b_2$ with

$$a_1 \in e_1 V e_2, \quad a_2 \in e_2 V e_3, \quad b_1 \in e_2 V e_1, \quad b_2 \in e_3 V e_2,$$

and let $A = T_S(V)/(R)$ with $R = \operatorname{span}\{a_1 b_1,\ b_2 a_2,\ b_1 a_1 - a_2 b_2\}$.

i. Check that

$$h_A(t) = \begin{pmatrix} 1 & t & t^2 \\ t & 1+t^2 & t \\ t^2 & t & 1 \end{pmatrix}.$$

ii. Show that $A_d = 0$ for $d \geq 3$.

iii. Using (a), (b) and Assignment 2, Exercise 3, part (2), prove that A is not Koszul.

(2) Let $S = \bigoplus_{x=1}^{n} \Bbbk e_x$, let V be a finite-dimensional S-bimodule in degree 1, and let $A = T_S(V)$. Prove that the multiplication map $V \otimes_S A \to A_+$ is an isomorphism of right A-modules, and deduce that S has the minimal graded projective resolution

$$0 \to (V \otimes_S A)(-1) \to A \to S \to 0,$$

so that A is Koszul.

(3) Let $S = \bigoplus_{i=1}^{n} \Bbbk e_i$, let V be a finite-dimensional S-bimodule in degree 1, put $C_{ij} = \dim_{\Bbbk} e_i V e_j$, and let $A = T_S(V)/(V \otimes_S V) = S \oplus V$.

i. Prove that $A^! = T_S(D(V))$.

ii. Prove that $h_A(t) = I + tC$ and $h_{A^!}(t) = (I - tC^{\mathsf{T}})^{-1}$.

iii. Deduce that $h_A(t)\, h_{A^!}(-t)^{\mathsf{T}} = I$.

iv. Prove that the minimal graded projective resolution of S has $P_h \cong (V^{\otimes_S h} \otimes_S A)(-h)$ for all $h \geq 0$.
Hint. Show that $\operatorname{Ker} d_0 = V$ and that $V A_+ = 0$, so each kernel is killed by A_+.

v. Deduce that A is Koszul.

(4) Let $A = \Bbbk[u, v]/(uv)$ with $\deg u = \deg v = 1$.

i. Prove that $ua + vb = 0$ in A if and only if $a \in vA$ and $b \in uA$.

ii. Show that the minimal graded projective resolution of $\Bbbk = A/A_+$ has $P_0 = A$ and $P_h \cong A(-h)^{\oplus 2}$ for all $h \geq 1$.

iii. Deduce that A is Koszul.